

Edge Detection Operators

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Introduction

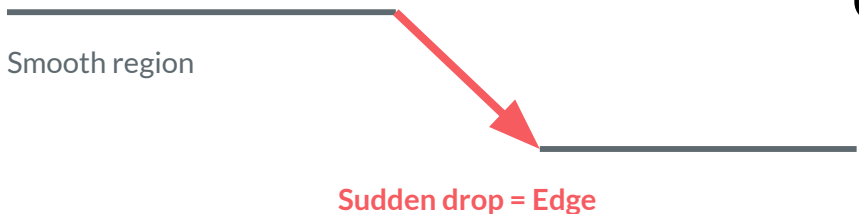
Introduction: What Edge Detection Finds

Intensity changes are like looking through fog:

Sharp brightness changes
mark object boundaries.

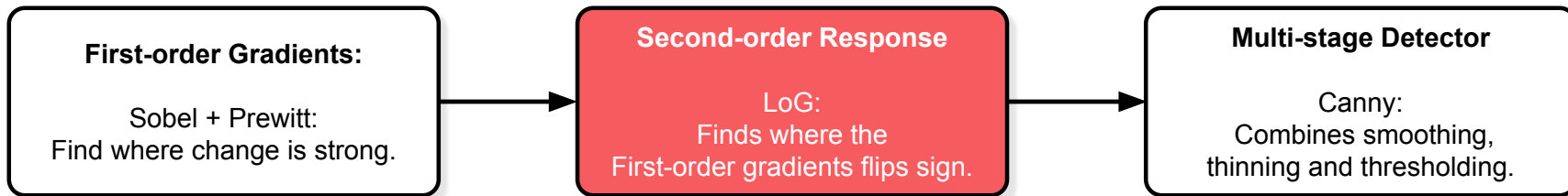
Edge detection reduces an image to its meaningful
intensity changes, while keeping shape
information.

Used before segmentation,
object extraction and recognition.



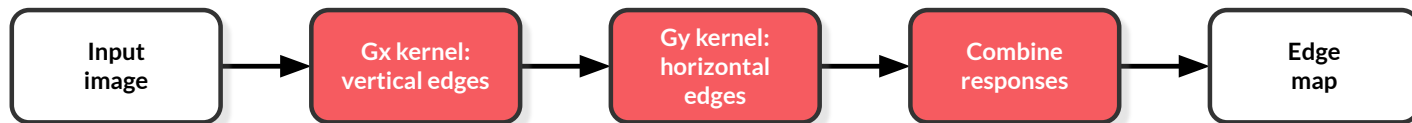
Introduction: Families of Edge Detectors

The operators in our presentation detect the same idea in different ways:



Sobel Operator

Sobel: Gradient-Based Edge Detection



Sobel is a first-order derivative method: it looks for large local intensity changes, not zero-crossings.

Gx		
-1	0	1
-2	0	2
-1	0	1

Gy		
1	2	1
0	0	0
-1	-2	-1

Gradient Magnitude:

$$\sqrt{(Gx^2 + Gy^2)}$$

$$\approx [Gx] + [Gy]$$

Edge Direction:

$$\tan^{-1}(Gy / Gx)$$

The centre row/column is weighted more strongly, so Sobel smooths slightly while differentiating.

Sobel: Worked Example

3×3 image patch

10	10	10
0	0	0
0	0	0

Gy kernel

1	2	1
0	0	0
-1	-2	-1

Gx kernel

-1	0	1
-2	0	2
-1	0	-1

$$(10 \times 1 + 10 \times 2 + 10 \times 1) + 0 + 0 = 40$$

$$(10 \times -1 + 10 \times 0 + 10 \times 1) + 0 + 0 = 0$$

Gy = 40

Gx = 0

Sobel: Worked Example (cont.)

Remember:

$$G_y = 40$$

$$G_x = 0$$

$$\text{Gradient Magnitude} = \sqrt{G_x^2 + G_y^2} = \sqrt{40^2 + 0^2} = 40$$

A high magnitude means Sobel has detected a strong intensity change.

$$\text{Edge direction} = \tan^{-1}(G_y / G_x) = \tan^{-1}(40 / 0) = \text{undefined}$$

This is a special case where the Edge direction is not computable.
This means that the edge line is perpendicular to the gradient direction
Therefore, in practice it is taken as a 90 degree gradient direction

Sobel: Strengths & Weaknesses

Strengths

Simple, fast and easy to implement.

Works well for strong outer boundaries.

Gives useful edge orientation from G_x and G_y .

Limitations / Failure Cases

Still sensitive to noise, weak contrast and heavy texture.

Can produce thick, broken or false edges.

Does not do edge thinning or linking like Canny.

Best use of Sobel Operator:

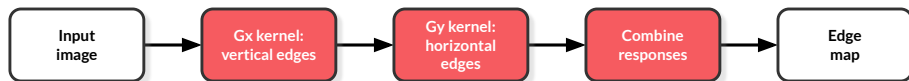
Fast & basic edge detection where a rough direction estimate is useful.

Prewitt Operator

Prewitt : Gradient-Based Edge Detection

Similarities to Sobel:

First-order derivative method:



Gradient Magnitude:

$$\sqrt{Gx^2 + Gy^2}$$

$$\approx [Gx] + [Gy]$$

Edge Direction:

$$\tan^{-1}(Gy / Gx)$$

Uniform weights across the kernel:

Gx

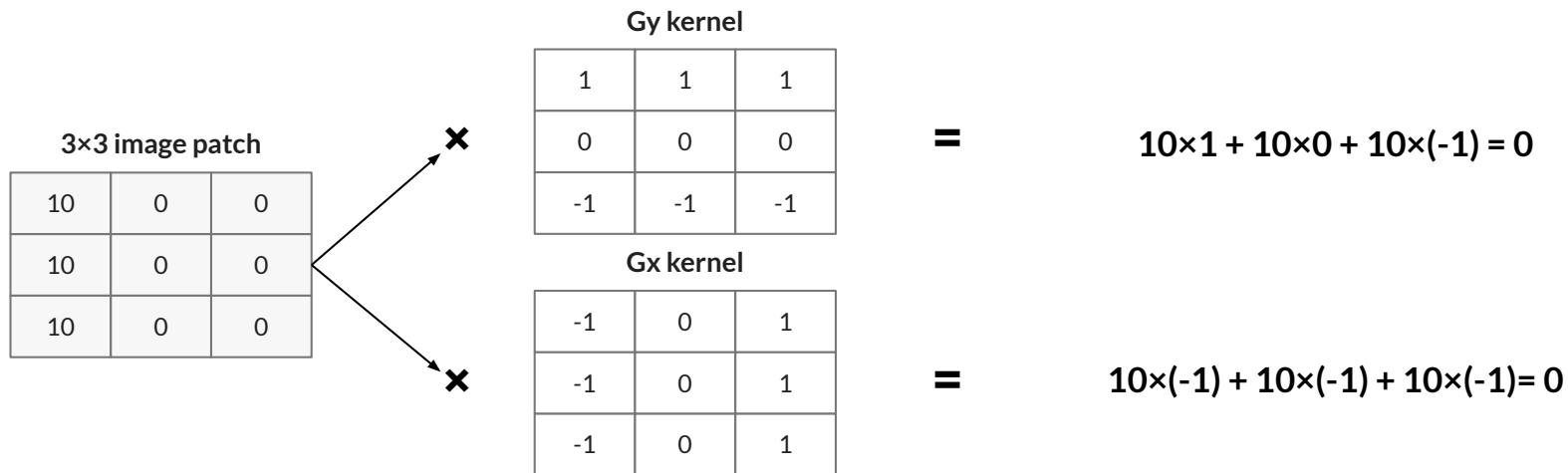
-1	0	1
-1	0	1
-1	0	1

Gy

1	1	1
0	0	0
-1	-1	-1

All kernel values are equally weighted, so Prewitt is an average-gradient detector rather than a smoothed one.

Prewitt: Worked Example



Gy = 0

Gx = -30

Prewitt: Worked Example

$$\text{Gradient Magnitude} = \sqrt{(G_x^2 + G_y^2)} = \sqrt{(0^2 + [-30]^2)} = 30$$

High magnitude: strong intensity change.

$$\text{Edge direction} = \tan^{-1}(G_y / G_x) = \tan^{-1}(0 / -30) = 0$$

Since $G_x < 0$, the gradient vector points entirely in the negative x-direction, so the actual direction is 180° . The gradient direction is perpendicular to the edge.

Gradient vector:



Edge detected: Vertical



Prewitt: Strengths & Weaknesses

Strengths

Easy to understand and implement.

Computationally light.

Works best on clean images with clean boundaries.

Limitations / Failure Cases

Weak on noisy image and subtle details.

Can miss fine structure and produce unstable edges.

LoG: Laplacian of Gaussian

LoG: Laplacian of Gaussian and How It Works



LoG is a second derivative method:

First it smooths the image with a Gaussian filter.

It then applies the Laplacian, which measures second-order intensity change.

Edges are detected where the filtered response crosses zero.

This differs from Sobel/Prewitt, which look for large first-order gradients.

LoG: Technical Method

Applying Laplacian to a Gaussian-smoothed image utilises the second derivative, as seen in the the formula:

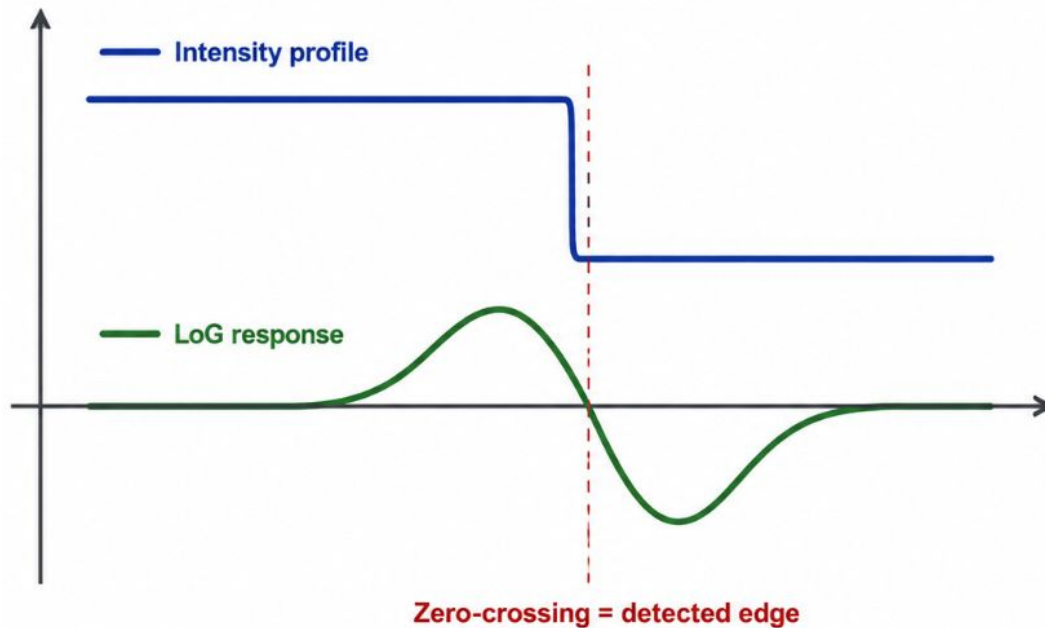
$$\text{LoG}(I) = \nabla^2(G_\sigma * I)$$

σ controls the amount of smoothing

larger σ = more smoothing, less detail and cleaner edges

smaller σ = less smoothing, more detail and more noise

LoG: Interpreting Results



Blue = original image intensity profile.

Green = result after Gaussian smoothing + Laplacian filtering.

LoG does not look for the highest gradient peak, unlike for instance Sobel, instead it looks for where the second derivative changes sign, meaning that the edge is detected at 0.

LoG: Strengths and Weaknesses

Strengths

Rotationally symmetric, so it does not favour only horizontal or vertical edges.

Able to detect contours clearly using zero-crossings.

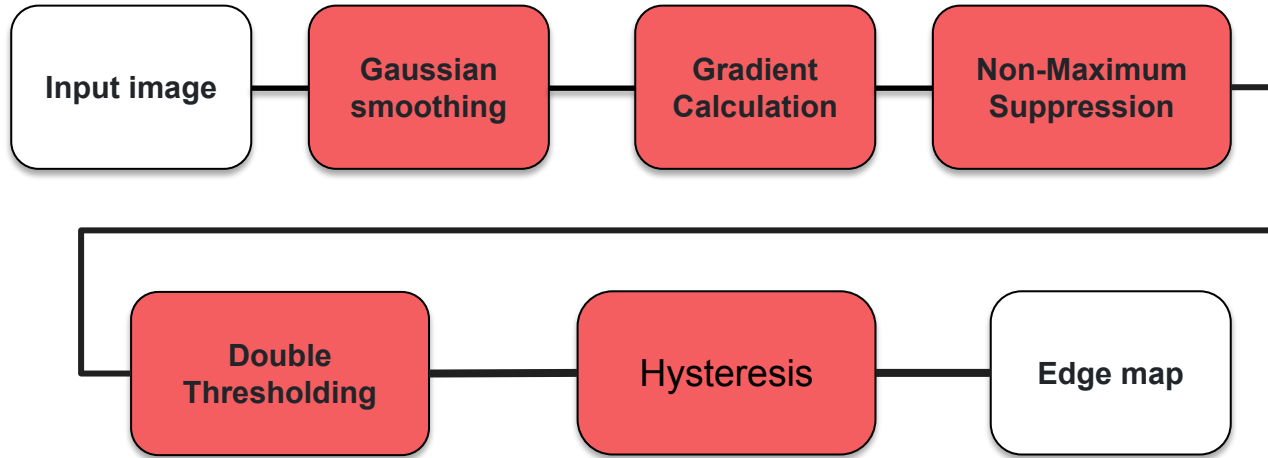
Limitations / Failure Cases

Due to using the 2nd derivative, does not directly give edge orientation like Sobel or Prewitt

Sensitive to Gaussian scale and thresholding choices.

Canny

Canny: Multi-Stage Edge Detection



- Canny uses a full pipeline instead of a single filter.
- It reduces noise, detects intensity changes, and thins edges.
- The final result is usually cleaner, thinner, and more continuous than Sobel or Prewitt.

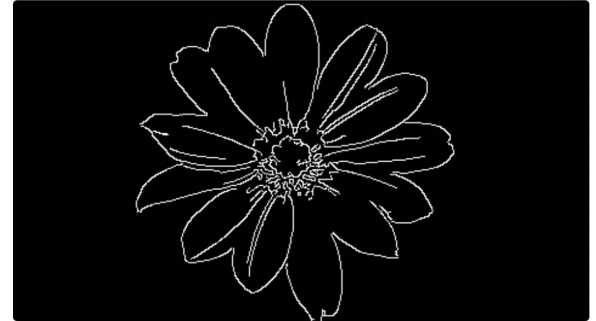
Canny Pipeline: How It Works



Original Image



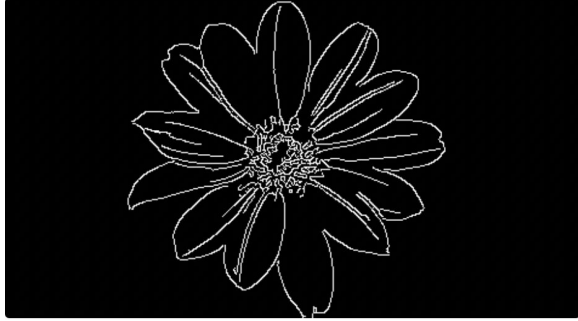
Smoothed Grayscale Image



Canny Output

Canny smooths the image to reduce noise, detects strong brightness changes, thins the detected edges, and keeps only connected reliable edges.

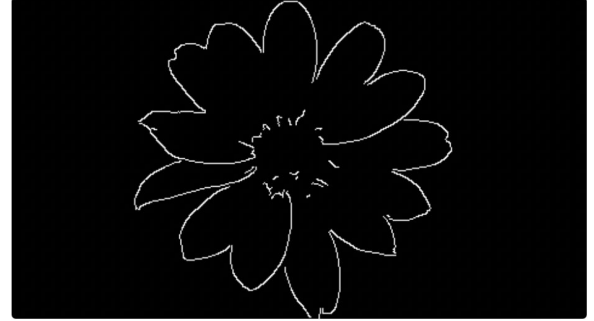
Canny Parameters and Sensitivity



Low Threshold: 20



Medium Threshold: 100



High Threshold: 200

As the threshold increases, Canny becomes stricter: noise and weak details are removed, but fine edges may also disappear.

Canny: Strengths and Weaknesses

Strengths

- Produces thin and clean edges.
- Reduces noise using Gaussian Smoothing.
- Keeps connected edges using hysteresis.
- Usually gives better edge maps than Sobel, Prewitt, and LoG.

Weaknesses

- More complex and slower than simple filters.
- Sensitive to threshold and blur settings.
- Can miss weak edges if thresholds are too high.
- Can still struggle with very noisy or low-contrast images.

Comparison And Real-World Example

Comparison

First-order gradient operators : Sobel and Prewitt

Easy and light

Gives edge orientation

Sensitive to noise, weak contrast, heavy texture...

Laplacian of Gaussian

Detects contour clearly

Not rotation dependent

Does not give orientation

Sensitive to noise, can produce double edges

Canny

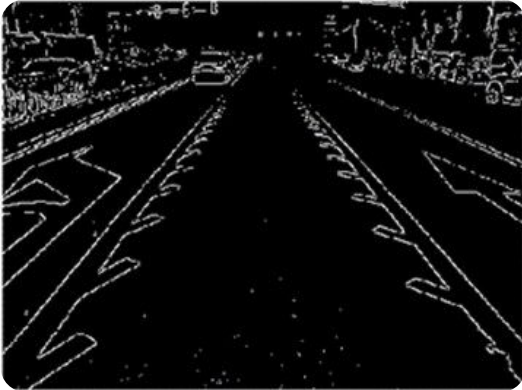
Smoothing, Gradients, Thinning, and thresholding

Clean, continuous edges

Complex and heavy

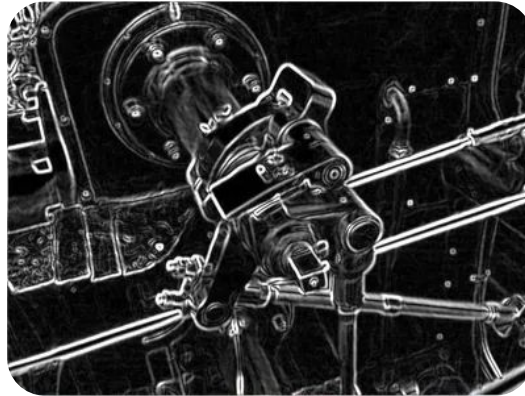
Comparison

First-order gradient operators : Sobel and Prewitt



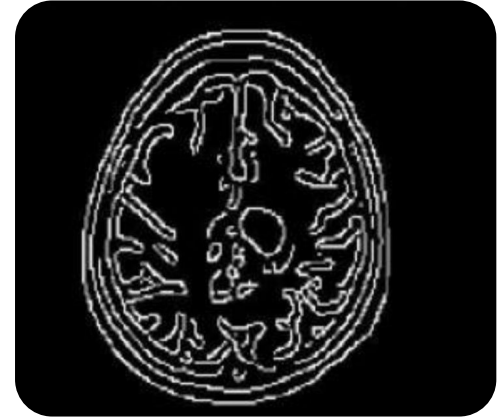
Best for clean lines, heavy contrast

Laplacian of Gaussian



Contour detection in any direction

Canny



Thin lines and clear end result

Quiz

Quiz: Testing Understanding

The quiz is designed to teach through common mistakes. Each question includes plausible wrong answers based on typical misunderstandings, so students have to reason through the concept rather than simply recall definitions.

Provides an understanding
on what each edge detection
operator does better
than the other

2. How does the Prewitt operator differ from the Sobel operator in its approach to detecting edges?

* 1 point

- ☒ It uses uniform weights across the kernel instead of giving extra weight to the centre pixel
- ☐ It uses second-order derivatives to find zero-crossings instead of first-order gradients
- ☐ It includes a mandatory Gaussian smoothing stage before any gradient calculations
- ☐ It is specifically designed to be more robust to noise than the Sobel operator